

# Improving Low-Resolution Face Recognition under Limited Data: How Synthetic Data Generation Can Close the Domain Gap

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IEEE International Joint  
Conference on Biometrics



FRONTX

## Introduction

- **Context:** Face Recognition (FR) in **surveillance** frequently meets **Low Resolution (LR)** probes, with faces spanning only 16–32 px after alignment, matched against a **High Resolution (HR)** gallery (mugshot, ID document). Naive upsampling does not restore identity.
- **Limited data:** labelled HR training data is abundant, but labelled *native*-LR identities, and above all *paired* native-LR/HR images, are scarce.
- **Common practice:** *synthesize* LR data by degrading HR faces, and validate on *synthetically* degraded benchmarks.
- **Question:** **Does a method trained on synthetic LR actually work on real LR?** And how much synthesis effort is repaid in accuracy for a compact, edge-oriented backbone?

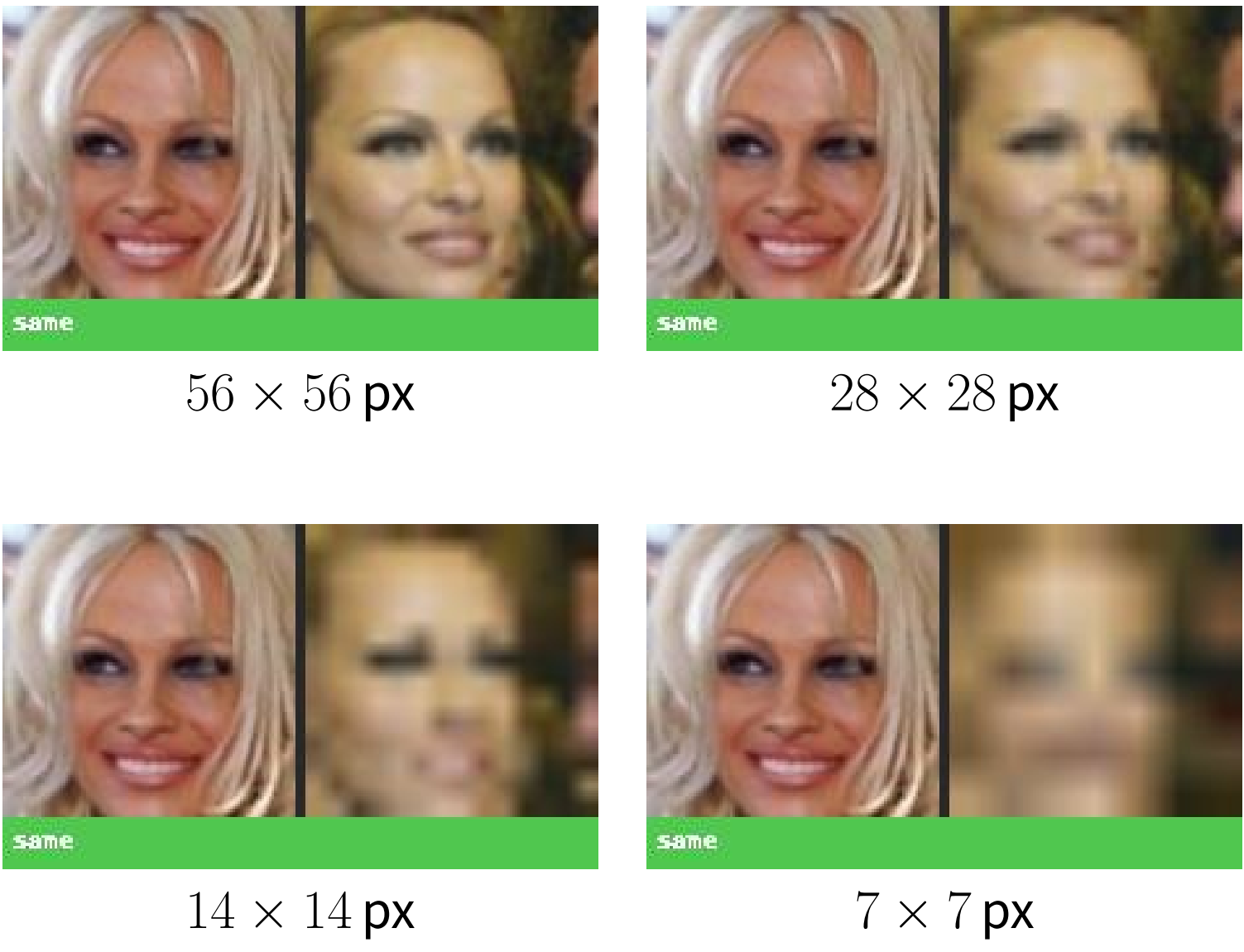


Figure: A verification pair at decreasing target resolution, each upsampled back to 112 px. Identity cues vanish progressively.

## Key contributions

- **Systematic study** of synthetic LR data generation for a compact backbone (EdgeFace-S) at three **synthesis efforts**: interpolation, Real-ESRGAN-style degradation, and a learned identity-aware super-resolution (SR) front-end.
- **Synthetic–real gap:** the same models evaluated on synthetic cross-resolution benchmarks (LFW, CFP-FP, AgeDB-30) *and* on real native-LR TinyFace; the optimal degradation differs.
- **Practical guidance:** match synthesis effort to the deployment setting; always report a *direct-feed* baseline and validate on *real* LR.
- **Fairness:** demographic bias assessment on RFW; accuracy gains do not reduce disparity.
- **Released pipeline** at [idiap.ch/paper/synth-lrfr](https://idiap.ch/paper/synth-lrfr).

## Synthesis strategies by effort

All LR training data are derived from HR WebFace4M. A setting is denoted “ $s \downarrow d / \uparrow u$ ”: downsample to  $s$  px with  $d$ , upsample to 112 px with  $u$  (e.g.  $28 \downarrow c / \uparrow a$  = cubic-down, area-up). **Low: interpolation augmentation.** Downsample to  $s \in \{56, 28, 14, 7\}$  with {area, cubic}, upsample back, and retrain EdgeFace-S end-to-end (CosFace, PartialFC, AdamW, 100 epochs). No HR data, no teacher, no learned component.

**Medium: generative degradation synthesis.** Ported Real-ESRGAN high-order degradation: random blur kernels, random resize, Gaussian/Poisson noise, JPEG cycles (two orders), final resize to  $32 \times 32$ . Also a *native* 32 px *stem*: EdgeFace-S with the stride-4 stem replaced by a  $1 \times 1$  conv, fed the 32px image directly.

**High: learned SR front-end.** Upsampling moves *inside* the model: two  $\times 2$  sub-pixel (PixelShuffle) stages  $32 \rightarrow 64 \rightarrow 128$  px, then four valid  $k=5$  convs to 112px; no interpolation. Feature blocks: plain conv (ESPCN, 0.86 M) or RRDB (Real-ESRGAN, 6.5 M). Prepend to a PDT translator in front of a *frozen* EdgeFace-base (Fig. 4).

- **Stage 1 (SR pretraining)**, identity-aware:
$$\mathcal{L}_{\text{SR}} = \ell_1(\text{SR}(x_{\text{LR}}), x_{\text{HR}}) + \lambda_{\text{id}}(1 - \cos(\phi(\text{SR}(x_{\text{LR}})), \phi(x_{\text{HR}})))$$
with  $\phi$  the frozen backbone embedding.
- **Stage 2 (joint contrastive):** upsampler + PDT trained with a contrastive loss over  $P \times K$  HR/LR batches; backbone frozen.

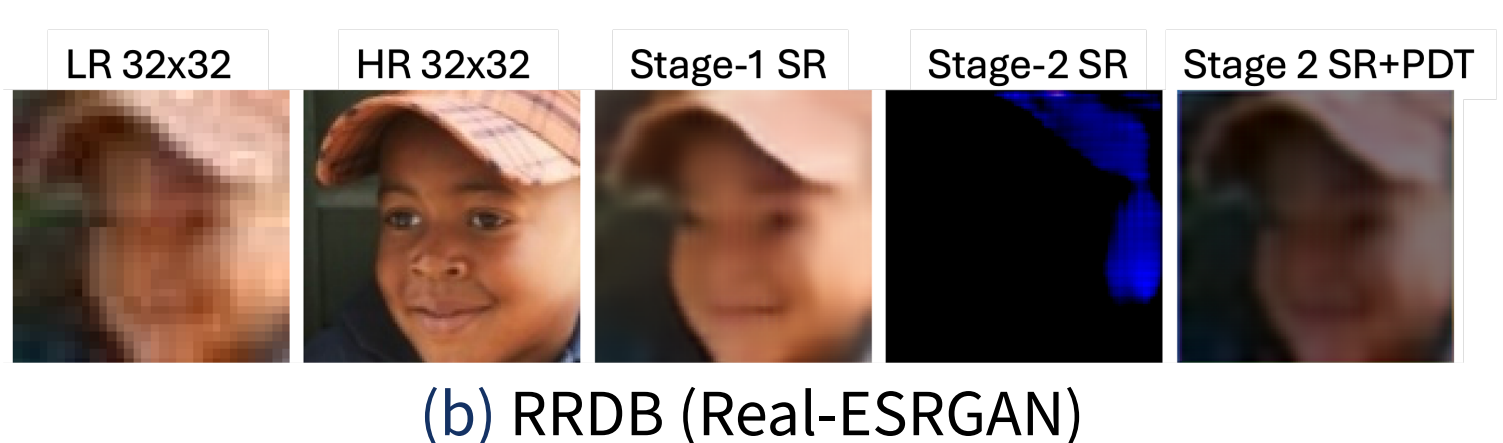


Figure: Learned 32 → 112 reconstructions on WebFace4M pairs after identity-aware pretraining (input / SR output / HR target). Plausible, identity-consistent detail is recovered, but this does not translate into recognition gains (Tab. 4).

## Baselines, Super Resolution, and domain adaptation

**Direct feed.** The aligned LR face is resized to the input size and passed to an HR-trained backbone (EdgeFace-S or base) with *no* front-end and no LR training: the simplest deployment and, as we show, a **hard baseline**.

**Knowledge distillation (KD).** A frozen HR-trained EdgeFace-S teacher guides an LR student of the same architecture with a CosFace loss plus embedding- and classifier-weight matching terms.

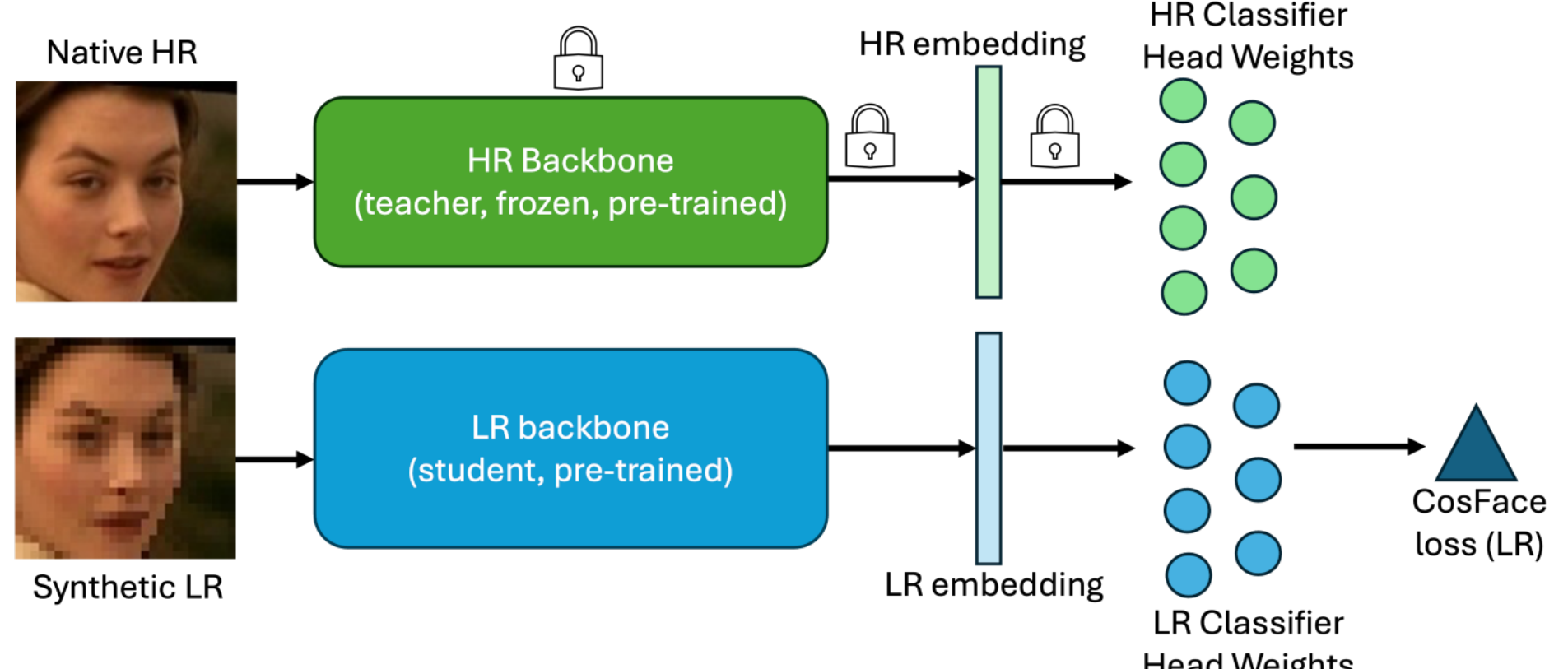


Figure: Adopted KD paradigm: the LR backbone is trained to produce embeddings similar to the HR backbone from LR data.

**Prepended Domain Transformer (PDT).** Input-side adaptation: the backbone is frozen and only a translator on the probe is trained on synthetic LR.

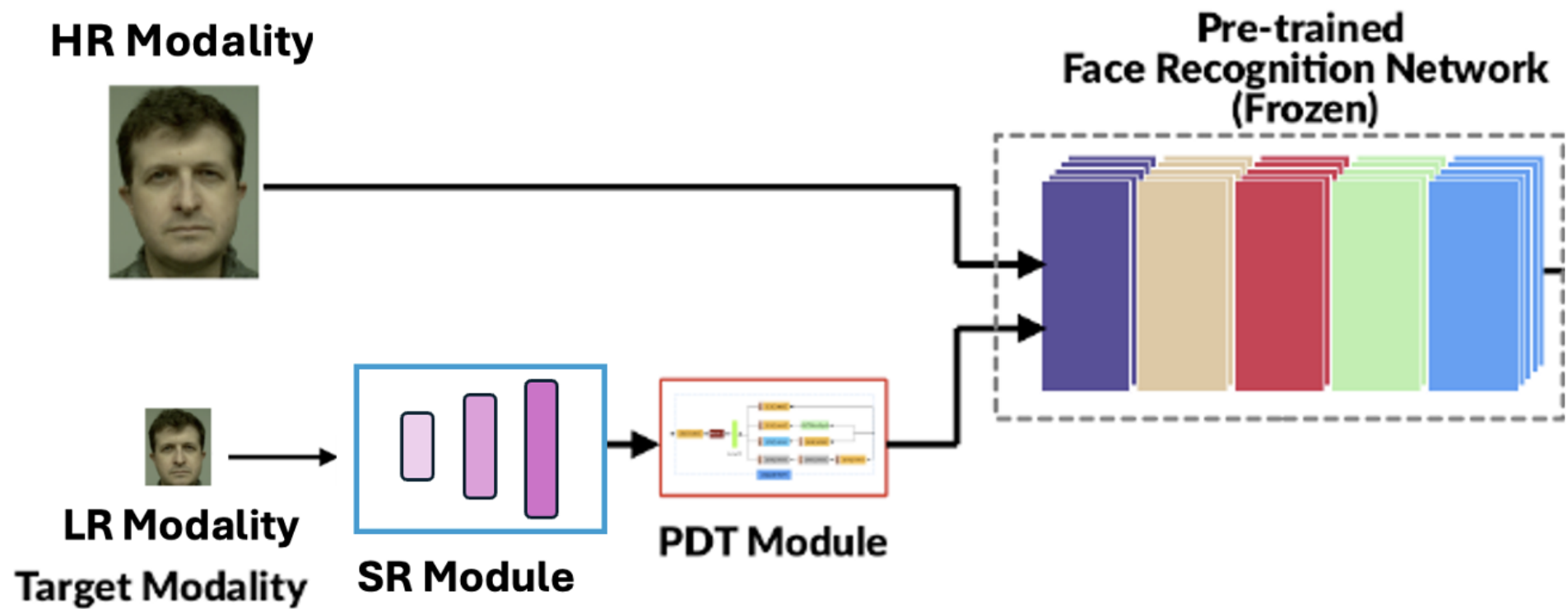


Figure: Input-side adaptation: a learned SR upsampler and a PDT translator map a 32 px native-LR face into the input domain of a frozen backbone. Only the upsampler is pre-trained (identity-aware, Stage 1), then upsampler and PDT are trained jointly with contrastive learning (Stage 2).

## The synthetic–real gap

**On synthetic cross-resolution benchmarks**,  $28 \downarrow c / \uparrow a$  is the best setting for *every* method under  $\text{HR} \rightarrow \text{LR}$  and KD gives the largest gains.  $\text{LR} \rightarrow \text{LR}$  (both faces degraded) is consistently harder than  $\text{HR} \rightarrow \text{LR}$ .

| Method                                | HR    | 56 $\downarrow c / \uparrow a$ | 28 $\downarrow a / \uparrow c$ | 28 $\downarrow c / \uparrow a$ | 14 $\downarrow a / \uparrow c$ |
|---------------------------------------|-------|--------------------------------|--------------------------------|--------------------------------|--------------------------------|
| <i>HR <math>\rightarrow</math> LR</i> |       |                                |                                |                                |                                |
| LR backbone                           | 78.85 | 80.25                          | 78.77                          | <b>80.74</b>                   | 79.67                          |
| KD                                    | 78.85 | 80.74                          | 78.60                          | <b>82.19</b>                   | 79.82                          |
| PDT                                   | 78.85 | 79.86                          | 80.35                          | <b>81.22</b>                   | 79.92                          |
| <i>LR <math>\rightarrow</math> LR</i> |       |                                |                                |                                |                                |
| LR backbone                           | 78.28 | 78.92                          | 78.74                          | <b>79.06</b>                   | 78.31                          |
| KD                                    | 78.28 | 79.14                          | 78.91                          | <b>80.59</b>                   | 78.51                          |
| PDT                                   | 78.28 | 78.81                          | <b>79.10</b>                   | 79.08                          | 78.54                          |

Table: Synthetic cross-resolution macro-mean accuracy (LFW, CFP-FP, AgeDB-30), by modality. Best LR-trained per row in **bold**.

**On real native LR the ranking flips.** The synthetic optimum  $28 \downarrow c / \uparrow a$  is the *worst* of the family on TinyFace, below the HR-trained model, while the milder  $56 \downarrow c / \uparrow a$  is the best.

| Evaluation            | HR-tr. | 56 $\downarrow c / \uparrow a$ | 28 $\downarrow c / \uparrow a$ | 14 $\downarrow a / \uparrow c$ |
|-----------------------|--------|--------------------------------|--------------------------------|--------------------------------|
| Synthetic macro (Acc) | 78.85  | 80.25                          | <b>80.74</b>                   | 79.67                          |
| Real TinyFace (mAP)   | 52.55  | <b>54.68</b>                   | 49.31                          | —                              |

Table: The same EdgeFace-S models, *synthetic* vs *real* LR. Synthetic:  $\text{HR} \rightarrow \text{LR}$  macro-mean Acc. Real: TinyFace mAP (DFA). The best resolution differs.

## Generative pipeline on real native-LR TinyFace

Medium- and high-effort synthesis on *real* native-LR TinyFace (DFA alignment); each model is compared against the *direct feed* of its own backbone, and the learned SR front-end is ablated with and without the joint PDT stage.

| Model                                 | mAP          | R-1          | R-5   | R-10  | R-20  |
|---------------------------------------|--------------|--------------|-------|-------|-------|
| base, direct (HR)                     | 65.12        | 71.08        | 74.97 | 75.83 | 76.98 |
| base + <i>RRDB-SR</i> + <i>PDT</i>    | <u>57.53</u> | 64.70        | 70.52 | 72.26 | 74.20 |
| base + <i>ESPCN-SR</i> + <i>PDT</i>   | 56.64        | 63.95        | 69.74 | 71.62 | 73.44 |
| S, HR-trained (direct)                | 52.55        | 59.66        | 66.63 | 69.05 | 71.49 |
| S, 56 $\downarrow c / \uparrow a$ aug | <b>54.68</b> | <b>61.40</b> | 68.16 | 70.28 | 72.80 |
| S, <i>Real-ESRGAN</i> 32 <i>stem</i>  | 53.74        | <b>61.40</b> | 68.62 | 70.84 | 73.47 |
| S, 56 $\downarrow c / \uparrow a$ PDT | 51.94        | 58.88        | 66.17 | 68.51 | 70.79 |
| S, 28 $\downarrow c / \uparrow a$ aug | 49.31        | 57.16        | 64.14 | 66.71 | 69.21 |

Table: TinyFace identification (DFA alignment,  $\times 100$ ). “base”=frozen EdgeFace-base (WebFace12M); “S”=EdgeFace-S (3.65 M); *direct* feeds the aligned LR face with no front-end. Generative methods in *italic*. **Bold**=best compact (S) model; underline=best short of the large direct-feed reference.

- **Compact backbone:** simple LR-aware augmentation ( $56 \downarrow c / \uparrow a$ ) is the *only* synthesis that beats its own direct feed, ahead of the Real-ESRGAN 32 px stem.
- **Strong frozen backbone:** the full SR+PDT pipeline trails a direct feed of the aligned LR image; SR alone is worse still.

## Does synthesis effort pay off? Not monotonically

- On the compact backbone the *lowest*-effort option beats both the no-synthesis baseline and the medium-effort degradation stem, while 28 px (optimal on synthetic data) is the worst. The *highest*-effort option is the best for *adaptation*-type approaches, but only on the large frozen backbone, and it is still outperformed by feeding that backbone directly.
- In general, we observed that **lower-effort synthesis show a better trade-off on a compact model** on face recognition accuracy for synthetic and real-world benchmarks.

- **Where does the gap come from?** The synthetic optimum being the worst on real LR under the *same* alignment implicates the *degradation model*, while alignment confounds absolute accuracy.
- At equal parameter budget and schedule,  $56 \downarrow c / \uparrow a$  augmentation outperforms the official EdgeFace-S checkpoint on Rank-1/-5: native-LR accuracy is driven by LR-aware synthesis and adaptation rather than by architecture.

## LR training as a regularizer on HR

Training on LR-augmented data at 56 and 28 px *matches or exceeds* the HR baseline on every standard benchmark and on IJB-C. The aggressive  $14 \downarrow a / \uparrow c$  is the exception; the native 32 px stem collapses on HR.

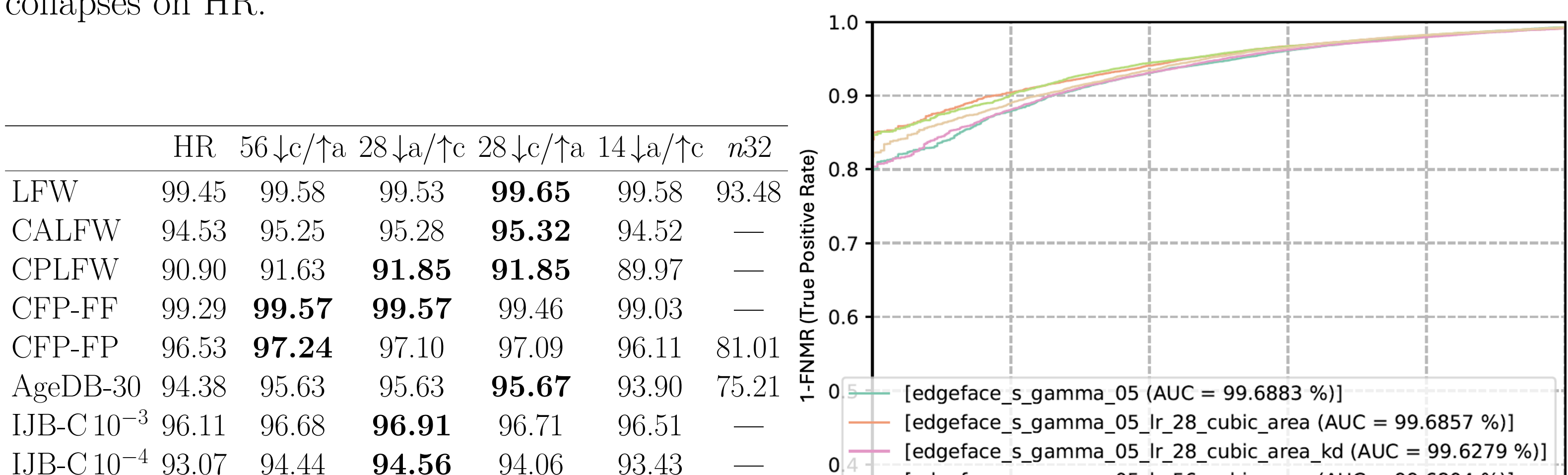


Figure: IJB-C FMR vs. 1-FNMR. The LR-trained EdgeFace-S variants match or exceed the HR baseline across operating points.

## Experimental setup

- **Training data:** WebFace4M in HR, its interpolation LR variants, and the Real-ESRGAN 32 px synthesis.
- **Backbones:** EdgeFace-S (3.65 M) for compact models; frozen EdgeFace-base (WebFace12M) for the SR+PDT front-end.
- **Synthetic benchmarks:** LFW, CFP-FP, AgeDB-30 degraded to 56/28/14/7 px, in  $\text{HR} \rightarrow \text{LR}$  (one image degraded) and  $\text{LR} \rightarrow \text{LR}$  (both) verification; standard HR sets and IJB-C TAR@FAR.
- **Real native LR:** TinyFace identification (mAP, Rank- $k$ ; gallery  $\sim 157k$ ) under **aligned\_pad** crops and the Differentiable Face Aligner (DFA, ResNet-50) from CVLface.
- Horizontal-flip test-time augmentation; all values in %.

## Bias and fairness on RFW

Per-group FMR on RFW (African, Asian, Caucasian, Indian) at the IJB-C threshold for  $\text{TMR@FMR} = 10^{-2}$ ; probes bicubically degraded to the test resolution.  $r_d = \text{FMR}_d / g$  with  $g$  the geometric mean over groups; Gini = 0 for equal FMR.

| px                                | Model  | African |             | Asian |      |
|-----------------------------------|--------|---------|-------------|-------|------|
|                                   |        | FMR     | $r$         | FMR   | $r$  |
|                                   | HR-tr. | 0.74    | <b>2.80</b> | 0.27  | 1.01 |
| HR 56 $\downarrow c / \uparrow a$ |        | 1.10    | <b>3.36</b> | 0.53  | 1.62 |
| 28 $\downarrow c / \uparrow a$    |        | 1.37    | <b>4.24</b> | 0.60  | 1.85 |
|                                   | HR-tr. | 0.77    | <b>3.41</b> | 0.43  | 1.92 |
| 56 56 $\downarrow c / \uparrow a$ |        | 0.90    | <b>3.25</b> | 0.57  | 2.04 |
| 28 $\downarrow c / \uparrow a$    |        | 1.31    | <b>3.85</b> | 0.43  | 1.28 |
|                                   | HR-tr. | 0.33    | <b>3.36</b> | 0.13  | 1.34 |
| 28 56 $\downarrow c / \uparrow a$ |        | 0.60    | <b>3.92</b> | 0.33  | 2.17 |
| 28 $\downarrow c / \uparrow a$    |        | 0.64    | <b>2.59</b> | 0.43  | 1.76 |

| px                                | Model  | Caucasian |             | Indian |      |
|-----------------------------------|--------|-----------|-------------|--------|------|
|                                   |        | FMR       | $r$         | FMR    | $r$  |
|                                   | HR-tr. | 0.07      | <b>0.25</b> | 0.37   | 1.39 |
| HR 56 $\downarrow c / \uparrow a$ |        | 0.03      | <b>0.10</b> | 0.60   | 1.82 |
| 28 $\downarrow c / \uparrow a$    |        | 0.00      | <b>0.05</b> | 0.80   | 2.47 |
|                                   | HR-tr. | 0.03      | <b>0.15</b> | 0.23   | 1.03 |
| 56 56 $\downarrow c / \uparrow a$ |        | 0.00      | <b>0.06</b> | 0.70   | 2.51 |
| 28 $\downarrow c / \uparrow a$    |        | 0.03      | <b>0.10</b> | 0.70   | 2.06 |
|                                   | HR-tr. | 0.00      | <b>0.17</b> | 0.13   | 1.33 |
| 28 56 $\downarrow c / \uparrow a$ |        | 0.00      | <b>0.11</b> | 0.17   | 1.08 |
| 28 $\downarrow c / \uparrow a$    |        | 0.03      | <b>0.14</b> | 0.40   | 1.62 |

Table: Per-group FMR (%) and ratio  $r$  to the geometric mean for different racial groups. px = test resolution; model = training regime. Gini coefficient for all groups.

- The Caucasian group is the most accurate ( $r \ll 1$ ) and the African group the least ( $r$  highest) in every cell, consistent with known RFW bias.
- LR-aware training does **not** systematically reduce the disparity: the Gini of the  $56 \downarrow c / \uparrow a$  and  $28 \downarrow c / \uparrow a$  backbones moves in both directions vs. HR-trained, and the largest group ratio belongs to  $28 \downarrow c / \uparrow a$ .

## Conclusions

- 1 **Synthetic  $\neq$  real:** the degradation optimal on synthetic benchmarks ( $28 \downarrow c / \uparrow a$ ) is the worst on real TinyFace, where the milder 56 px setting wins.
- 2 **Effort is not monotonic:** on a retrainable compact backbone, the cheapest interpolation augmentation outperforms Real-ESRGAN-style degradation and a learned SR front-end.
- 3 **Direct feed is a hard baseline:** an identity-aware SR front-end never beats feeding a strong frozen backbone the aligned LR image; report it before claiming any restoration/translation pipeline helps.
- 4 **Fairness:** accuracy gains from LR-aware synthesis do not reduce demographic bias on RFW.

**Recommendation:** match synthesis effort to the deployment constraint and **validate on real native LR**, not on synthetic degradation alone.

**Future work:** SR front-end on a *compact* backbone; pre-training on synthetic LR and fine-tuning on limited real LR (e.g. TinyFace train split); learning the paired HR-reference/LR-probe conditional directly (conditional/diffusion generators); fairness of LR capture.

## Code, paper and acknowledgements



<https://idiap.ch/paper/synth-lrfr/>

Funded by Frontex Research Grants Programme, Call 2024/CFP/INNOVATE/01, GA No. 2025/280; CarMen (HORIZON-CL3-2023-BM-01, No. 101168325); PopEye (HORIZON-CL3-2023-BM-01, No. 101168317); and CERTAIN (HORIZON-CL4-2024-DATA-01, No. 101189650).